

Name:	
Last Name:	
Student number:	

Please Note:

- The exam consists of **20 multiple-choice** and **10 True/False** questions. You must select the correct answer for each question and **circle** it on the provided question sheet.
- Additionally, there are **4 written** questions that require clear and concise answers in the designated space.
- The use of **electronic devices** (e.g., calculators, laptops, smartphones, tablets) is **strictly prohibited**.
- The exam duration is **75 minutes**. You are not allowed to leave the examination room within the first **15 minutes**.
- Please ensure you **sign** the attendance form. Any exam papers **without an attendance signature will not be considered**.

Good Luck!**Abbas Akkasi**

(Each True/False and multiple-choice question is worth 3 marks)

- 1- Feature selection and feature extraction are identical techniques, both aiming to reduce dimensionality by creating new synthetic features.
True False
- 2- Equal-frequency binning involves dividing the range of a continuous feature into intervals of equal width.
True False
- 3- In an SVM, the hyperplane is chosen to minimize the distance between itself and the closest data points from each class.
True False
- 4- The kernel trick allows SVMs to operate in a high-dimensional feature space without the computational cost of explicitly transforming the data into that space.
True False
- 5- In a convolutional layer, the values within a filter (or kernel) are pre-defined constants, such as the Sobel filter, and cannot be learned during training.
True False
- 6- Backpropagation Through Time (BPTT) works by unrolling the RNN, applying the standard backpropagation algorithm to each time step independently, and then updating the weights for each time step separately.
True False
- 7- Conditional independence means that if two variables, X and Y, are independent, they remain independent even after conditioning on a third variable Z.
True False
- 8- Naïve Bayes is considered robust to irrelevant features because the probability estimates for irrelevant features will roughly cancel out across classes.
True False

9- Different clustering algorithms applied to the same dataset will always produce identical results.

True False

10- The MIN (single-link) hierarchical clustering method defines the distance between two clusters as the distance between their farthest points.

True False

- 11- A data scientist wants to convert a continuous feature, 'Age', into a categorical feature with three categories: 'Young', 'Middle-aged', and 'Senior'. This process is an example of:**
- a) Standardization
 - b) Normalization
 - c) Binning
 - d) Age is not continues feature
- 12- After performing PCA on a dataset, you get the following eigenvalues for the 4 principal components: [3.2, 1.1, 0.4, 0.1]. What is the proportion of variance explained by the first principal component?**
- a) 3.2%
 - b) 32%
 - c) Approximately 67%
 - d) Approximately 80%
- 13- A key similarity between PCA and LDA is that both:**
- a) Are unsupervised learning algorithms.
 - b) Rank the new axes they create in order of importance.
 - c) Aim to maximize class separability.
 - d) Can produce a number of components equal to the number of original features.
- 14- What is the primary objective of a Maximum Margin Classifier (Hard Margin SVM)?**
- a) To minimize the number of support vectors.
 - b) To find any hyperplane that separates the classes.
 - c) To find a hyperplane that maximizes the distance to the closest points from each class.
 - d) To minimize the sum of squared errors on the training data.
- 15- Which of the following is stated as a primary advantage of SVM?**
- a) It performs exceptionally well on very large datasets with millions of samples.
 - b) It is highly interpretable, like a single decision tree.
 - c) It is memory efficient and effective in high-dimensional spaces.
 - d) It requires no parameter tuning.

- 16- For a data point x that lies exactly on the decision hyperplane, what is the value of $w \cdot x + b$?
- a) 1
 - b) -1
 - c) 0
 - d) It depends on the class of the point.
- 17- Why is a standard Multi-Layer Perceptron (MLP) generally not used for direct processing of raw, high-resolution images?
- a) MLPs cannot use non-linear activation functions.
 - b) The number of parameters becomes astronomically large, leading to overfitting and high computational cost.
 - c) MLPs require sequential data as input, which images are not.
 - d) MLPs are only capable of binary classification.
- 18- If you have an input volume of $32 \times 32 \times 3$ and use 10 filters of size 5×5 with stride 1 and 'same' padding, what is the spatial dimension (height and width) and depth of the output volume?
- a) $28 \times 28 \times 10$
 - b) $32 \times 32 \times 3$
 - c) $32 \times 32 \times 10$
 - d) $28 \times 28 \times 3$
- 19- The filters in the first layer of a trained CNN are most likely to detect:
- a) High-level concepts like faces or objects.
 - b) Low-level features like edges, colors, and blobs.
 - c) Complex patterns like eyes or wheels.
 - d) The class label of the image.
- 20- What is the primary benefit of weight sharing in Recurrent Neural Networks?
- a) It makes the network deeper and thus more powerful.
 - b) It allows the network to be trained with Backpropagation Through Time.
 - c) It enables the network to process sequences of variable length and generalize patterns across different positions in the sequence.
 - d) It completely eliminates the vanishing gradient problem.

- 21- The vanishing gradient problem in RNNs is most directly analogous to what phenomenon in a simple numerical product?**
- a) Multiplying many numbers that are all close to 1.
 - b) Multiplying many numbers that are all less than 1 (e.g., $0.1 * 0.1 * 0.1 \dots$).
 - c) Adding many numbers that are all greater than 1.
 - d) Dividing a large number by a very small number.
- 22- The fundamental recurrence relation for the hidden state h_t in a simple RNN is given by:**
- a) $h_t = f(W_x * x_t)$
 - b) $h_t = f(W_x * x_t + W_y * h_{t-1})$
 - c) $h_t = f(W_y * h_{t-1})$
 - d) $h_t = W_x * x_t + W_y * h_{t-1}$
- 23- Which formula correctly represents Bayes' Theorem?**
- a) $P(Y | X) = P(X | Y) \times P(Y)$
 - b) $P(Y | X) = \frac{P(X|Y)}{P(Y)}$
 - c) $P(Y | X) = \frac{P(X|Y)P(Y)}{P(X)}$
 - d) $P(Y | X) = \frac{P(X)P(Y)}{P(X|Y)}$
- 24- The Laplace estimate for a categorical feature addresses the zero-frequency problem by:**
- a) Discarding features that have zero probabilities.
 - b) Artificially increasing the count of each class outcome by one.
 - c) Replacing all zero probabilities with a very small constant like $1e-9$.
 - d) Increasing the count of each feature value for a given class by one and adjusting the denominator accordingly.
- 25- Which of the following is NOT a stated advantage or characteristic of the Naïve Bayes classifier?**
- a) It is robust to isolated noise points.
 - b) It performs well even when features are highly correlated and redundant.
 - c) It can handle missing values by ignoring them during probability estimation.
 - d) It is robust to irrelevant attributes.

- 26- You have a dataset where a categorical feature has a value that was never seen in the training data for a particular class. Without any form of smoothing, how will Naïve Bayes handle a test instance with this unseen feature value?
- a) It will ignore that feature for that instance and base the decision on the other features.
 - b) It will assign a very small probability to that feature for that class.
 - c) It will assign a probability of zero to that class for that instance.
 - d) It will throw an error and fail to classify the instance.
- 27- In which type of clustering is a cluster defined as a set of points such that any point in the cluster is closer to the "center" of that cluster than to the center of any other cluster?
- a) Well-separated clusters
 - b) Contiguity-based clusters
 - c) Prototype-based clusters
 - d) Density-based clusters
- 28- What does the 'd' represent in the K-means complexity formula $O(n * K * I * d)$?
- a) The distance between clusters.
 - b) The density of the data points.
 - c) The number of attributes (dimensions) of the data.
 - d) The number of iterations until convergence.
- 29- Which of the following is a major advantage of hierarchical clustering over partitional methods like K-means?
- a) It is much faster and can handle millions of data points easily.
 - b) It does not require the user to pre-specify the number of clusters.
 - c) It is guaranteed to find the globally optimal clustering.
 - d) All of the above options
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- 30- In DBSCAN, what is a "border point"?
- a) A point with fewer than MinPts neighbors within ϵ , but which is within the ϵ -neighborhood of a core point.
 - b) Any point that is not a core point.
 - c) A point with more than MinPts neighbors within ϵ .
 - d) A point that lies exactly on the boundary of the dataset's convex hull.
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- 31- You are designing a CNN for a small image classification task. The input images are **32x32 RGB**. You have a convolutional layer with **6 filters**, each of size **5x5**, a stride of **1**, and you choose to use **'same' padding**.
- a)** What are the dimensions (height, width, depth) of the output feature map from this layer? (6 marks)
- b)** If you then apply a **max-pooling** layer with a **2x2 filter** and a **stride of 2**, what are the dimensions of the resulting feature map? (6 marks)
- c)** If the output from the pooling layer is then flattened and fed into a **fully connected** layer with **100 neurons**, how many parameters (weights) are there between the flattened layer and this fully connected layer? (6 marks)

- 32- You are tasked with building a deep learning model for two different tasks:
- **Task A:** Predicting the daily closing price of a stock for the next 5 days, given its closing prices for the previous 60 days.
 - **Task B:** Classifying the sentiment of a movie review as either positive or negative.

For each task, answer the following:

a) Is this a **one-to-one, many-to-one, one-to-many, or many-to-many problem**? Justify your answer. (6 marks)

b) Which loss function would be the most appropriate to train your model? (6 marks)

- 33- Consider four nodes: "Flu" (F), "Fever" (E), "Cough" (C), and "Muscle Ache" (M). Let "Flu" be the root node, and let it directly influence "Fever" and "Cough". "Muscle Ache" is influenced by both "Fever" and "Cough".
- a)** Draw the DAG representing these relationships. (6 marks)
 - b)** Write the equation for the full joint probability $P(F, E, C, M)$ based on your network structure. (6 marks)

34- You are given the following one-dimensional data points: {1, 2, 5, 9}

You want to perform agglomerative hierarchical clustering using the MAX Link method.

a) Draw a table showing the initial distance matrix. (6 marks)

Use Euclidean distance (which in one dimension is simply the absolute difference).

b) Perform the first merge. Which two points are merged first? What is the distance value for this merge? (6 marks)

c) After the first merge, you have a new cluster. Update the distance matrix for the next step. Show the new distances between the new cluster and the remaining points. (6 marks)

You only need to show the updated row and column for the new cluster.